

BEEE - UNIT 2

Short Questions:

1. State the working principle of Three phase Induction Motor.

Ans: A three-phase induction motor works on the principle of **electromagnetic induction**.

When a three-phase AC supply is given to the stator, it produces a rotating magnetic field (RMF). This rotating magnetic field cuts the rotor conductors and induces an EMF in them. Since the rotor circuit is closed, current flows in the rotor.

According to Lenz's law, the interaction between the stator magnetic field and rotor current produces a torque, causing the rotor to rotate in the same direction as the rotating magnetic field.

2. Write basic working principle of DC- Generator?

Ans: A DC generator works on Faraday's law of electromagnetic induction: Whenever a conductor cuts magnetic flux, an EMF is induced in it. Mechanical energy supplied to the armature is converted into electrical energy.

3. Discuss the applications of induction motors.

Ans:

1. Pumps and compressors
2. Fans and blowers
3. Conveyors
4. Machine tools (lathe, drilling, milling)
5. Elevators and cranes
6. Domestic appliances (washing machines, mixers)

4. Discuss the working principle of DC Motor.

Ans: A DC motor works on Lorentz force principle: When a current-carrying conductor is placed in a magnetic field, it experiences a force. The interaction between armature current and field flux produces torque, converting electrical energy into mechanical energy.

Long Questions:

1. Explain the Construction and Operation of DC Generator.

Ans: Construction of DC Machine: A DC generator can be used as a DC motor without any constructional changes and vice versa is also possible. Thus, a DC generator or a DC motor can be broadly termed as a DC machine.

- ✓ A dc generator is an electrical machine which converts mechanical energy into electrical energy.
- ✓ A dc motor is an electrical machine which converts electrical energy into mechanical energy.

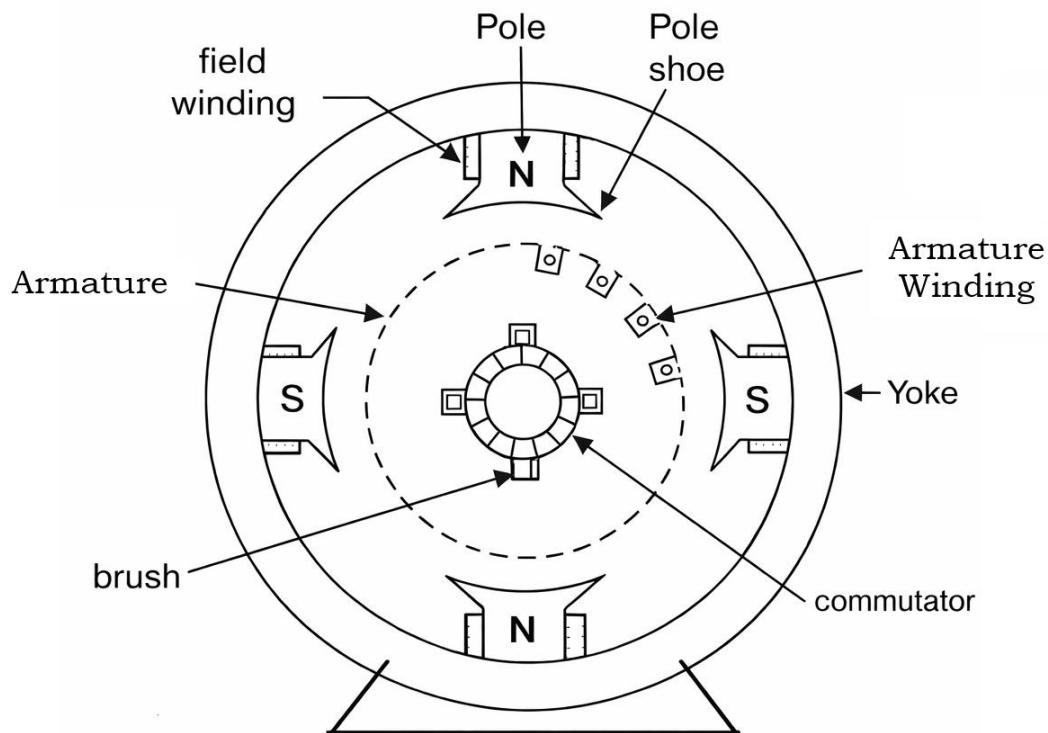


Fig: DC machine

The above figure shows constructional details of a simple 4-pole DC machine. A DC machine consists of two basic parts; stator and rotor.

Basic constructional parts of a DC machine are described below.

Stator Parts:

1. Magnet frame (or) Yoke
2. Pole and pole shoe
3. Field coils
4. Brushes
5. Brush Holders.

Rotor Parts:

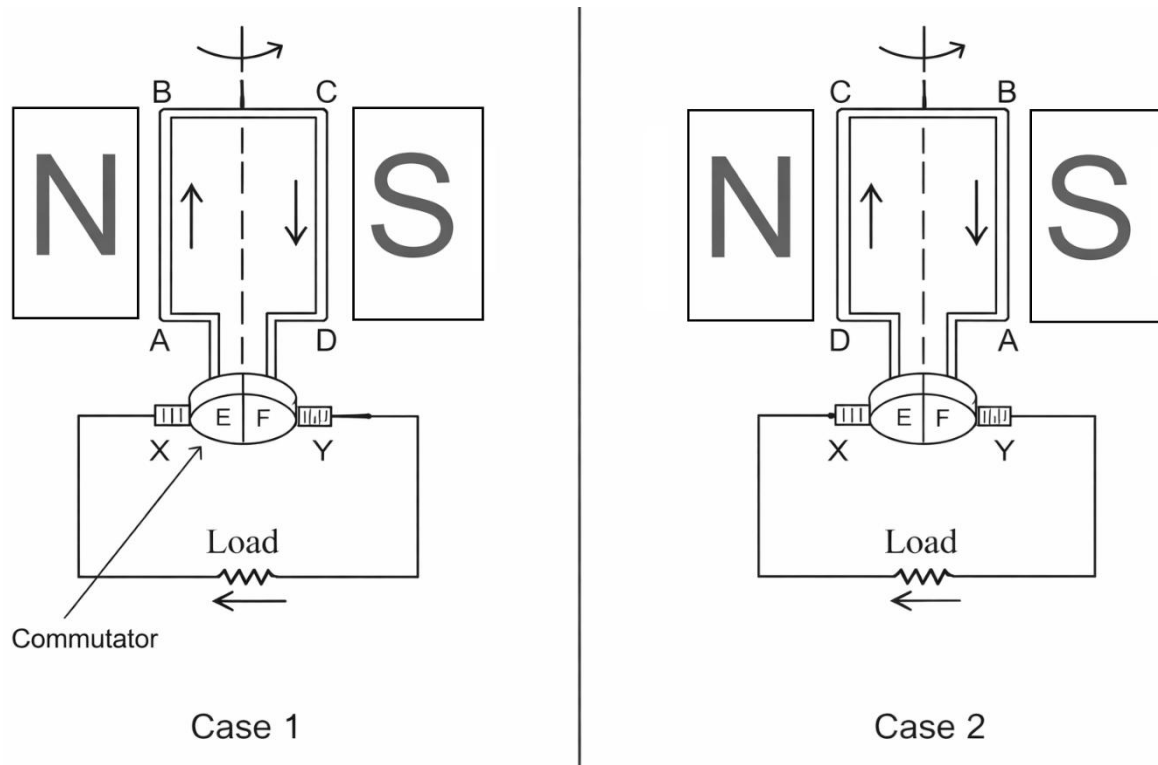
1. Armature Core 2. Armature Winding 3. Commutator

Parts of the DC Machine	Material made of	Function of parts
Magnetic frame or Yoke	cast iron or steel	1. It provides mechanical support to the whole assembly. 2. Carries the magnetic flux produced by the field winding.
Poles and pole shoes	cast iron or steel	1. They support field coils. 2. Spread out the flux in air gap uniformly.
Field coils	copper	Field coils are former wound and placed on each pole and connected in series. When energized, they form alternate North and South poles.
Armature core	Laminated circular steel disks	It is cylindrical in shape with slots to carry armature winding.
Armature winding	copper	It is usually a former wound copper coil which rests in armature slots.
Commutator	Copper	1. In case of DC generator, it converts AC to DC. 2. In case of DC motor, it converts DC to AC.
Brushes	Carbon or Graphite	1. In DC generator: lead the current to external load circuit from rotating armature. 2. In DC motor: lead the current to rotating armature.

2.Explain the working principle of DC Generator.

Ans: Working Principle of A DC Generator:

- According to Faraday's laws of electromagnetic induction, whenever a conductor is placed in a varying magnetic field (OR a conductor is moved in a magnetic field), an emf gets induced in the conductor.
- In a DC generator, field coils produce an electromagnetic field and the armature conductors are rotated into the field. Thus, an electromagnetically induced emf is generated in the armature conductors. The direction of induced current is given by Fleming's right hand rule.



- According to Fleming's right-hand rule, the direction of induced current changes whenever the direction of motion of the conductor changes.
- Let's consider an armature rotating clockwise and a conductor at the left is moving upward. When the armature completes a half rotation, the direction of motion of that particular conductor will be reversed to downward. Hence, the direction of current in every armature conductor will be alternating. If you look at the above figure, you will know how the direction of the induced current is alternating in an armature conductor.

- But with a split ring commutator, connections of the armature conductors also get reversed when the current reversal occurs. And therefore, we get unidirectional current at the terminals.

3. Explain the working principle of DC Motor.

Ans: Working principle of DC Motor:

- A DC motor is an electrical machine that converts electrical energy into mechanical energy.
- It works on the principle that whenever a current-carrying conductor is placed in a magnetic field, it experiences a mechanical force.
- The direction of the force is determined by Fleming's Left-Hand Rule.
- The magnitude of the force acting on the conductor is given by:

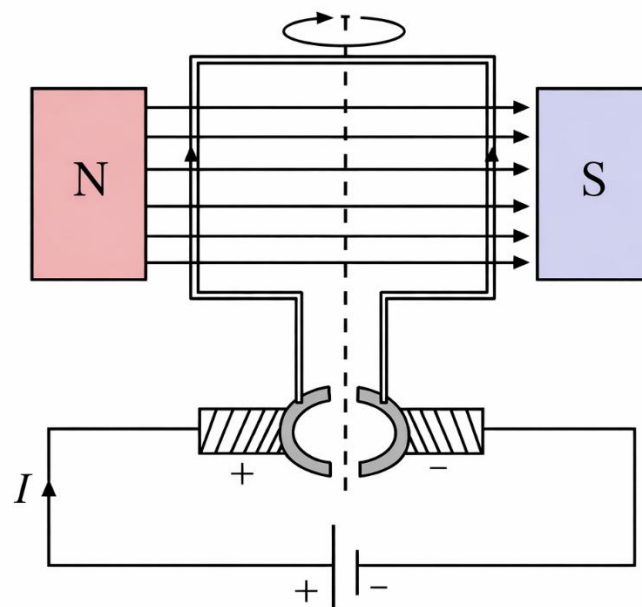
$$F = BIL$$

Where:

B = Magnetic flux density

I = Current in the conductor

L = Length of the conductor within the magnetic field.



Working Principle of DC Motor

- When the armature winding is connected to a DC supply, current flows through the armature conductors.

- The magnetic field is produced either by: Field winding (electromagnetism), or Permanent magnets.
- The interaction between the magnetic field and current-carrying conductors produces a force.
- Forces acting on opposite sides of the armature create a torque, causing the armature to rotate.
- Continuous rotation results in the conversion of electrical energy into mechanical energy.

4. **Explain the construction of a single-phase transformer.**

Ans: There are two basic parts of a transformer

- i) Magnetic Core
- ii) Winding or Coils.

Magnetic Core: Core is made up of silicon steel and it is laminated (0.3 to 0.5 mm thick). Because of laminated type of construction, eddy current losses get minimized. For this generally 'L' shaped or 'I' shaped laminations are used which are shown in the Fig 1.

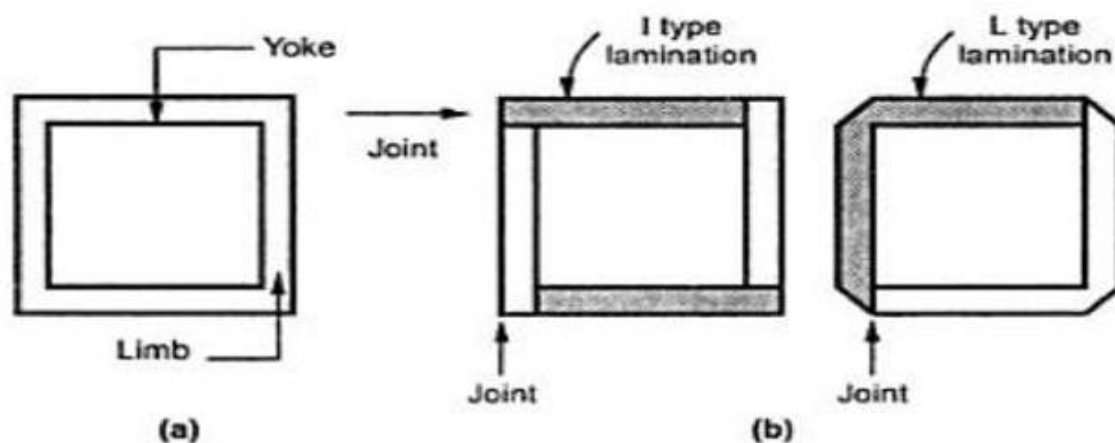


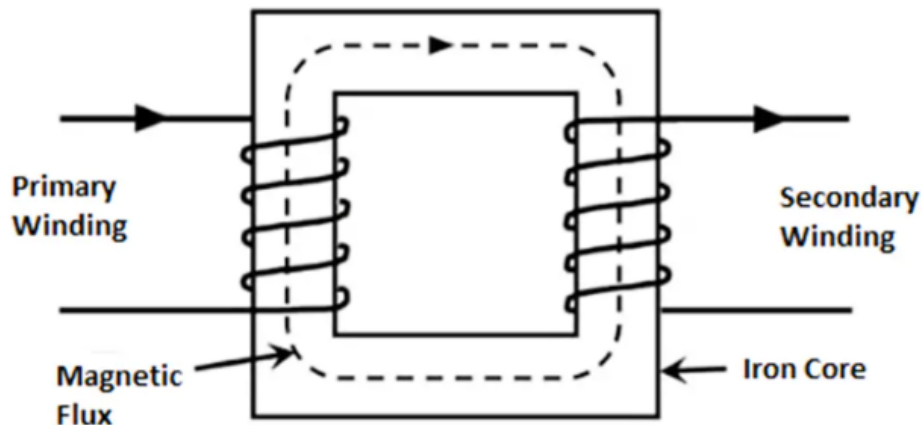
Fig. 1 Construction of transformer

Windings: The coils used are wound on the limbs and are insulated from each other. In the basic transformer, the two windings wound on two different limbs i.e. primary on one limb while secondary on other limb.

The various types of single-phase transformers are,

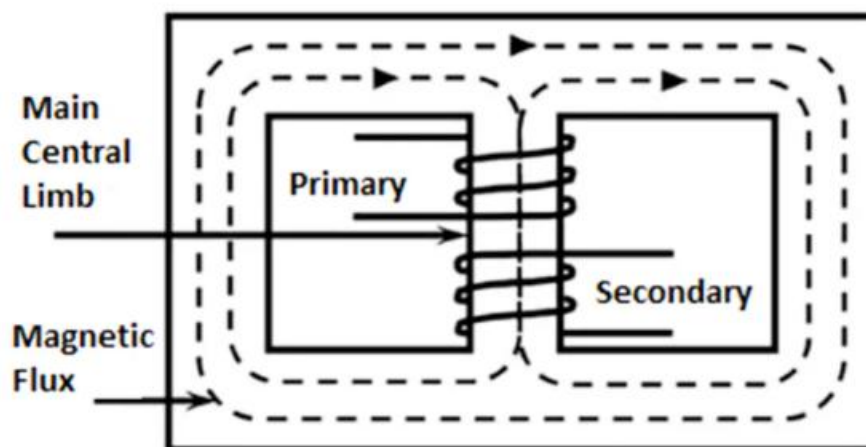
1. Core type
2. Shell type

Core Type Transformer: It has a single magnetic circuit. The rectangular core having two limbs. The core is surrounded by the winding. In this Cylindrical type coils are used. Both the coils are placed on both the limbs.



Core Type Transformer

2. Shell Type Transformer: It has a double magnetic circuit. The core has three limbs. Both the windings are placed on the central limb. The winding is surrounded by core. The coils used are generally multilayer disc type or sandwich coils. Generally, for very high voltage transformers, the shell type construction is preferred.



Shell Type Transformer

5. Explain the Principle and operation of Single - Phase transformer with a neat sketch.

Ans:

The working principle of transformer is **Faraday's law of electromagnetic induction**. Actually, mutual induction between two winding is responsible for transformation action in an electrical transformer.

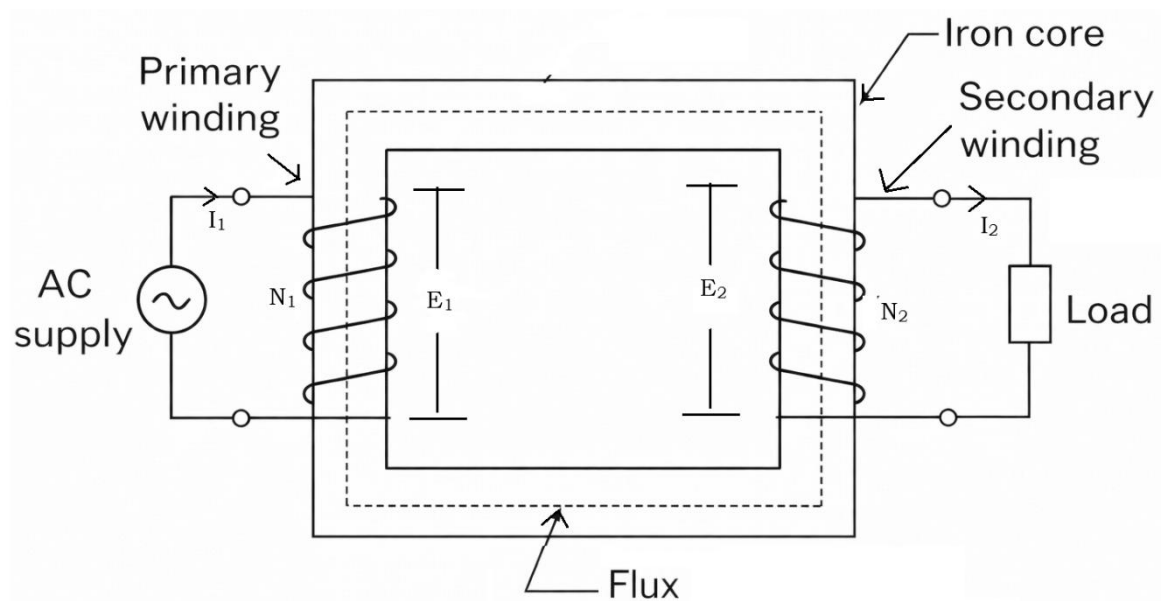


Fig (a): Single-Phase transformer.

- One of the two coils which one is connected to the alternating voltage is called primary winding. The other winding which is connected to load is called secondary winding.
- The primary winding has N_1 number of turns while the secondary winding has N_2 number of turns.
- When primary winding is excited by an alternating voltage, it circulates an alternating current. This current produces an alternating flux (Φ) which completes its path through common magnetic core as shown dotted in the Fig.(a). Thus an alternating flux links with the secondary winding.
- As the flux is alternating, according to Faraday's law of an electromagnetic induction, mutually induced e.m.f. gets developed in the secondary

winding. If now load is connected to the secondary winding, this e.m.f. drives a current through it.

- The amount of induced emf in each winding depends on the rate of change of magnetic flux and the number of turns in the coil. The induced emf in the primary (E_1) is proportional to its turns (N_1), and the emf in the secondary (E_2) is proportional to its turns (N_2).
- When $N_2 > N_1$, the transformer is called a step-up transformer, when $N_2 < N_1$ the transformer is called step-down transformer.

EMF equation of transformer:

$$\begin{aligned} E_1 &= 4.44 f \phi_m N_1 \\ E_2 &= 4.44 f \phi_m N_2 \end{aligned}$$

6. Explain the construction of induction motor.

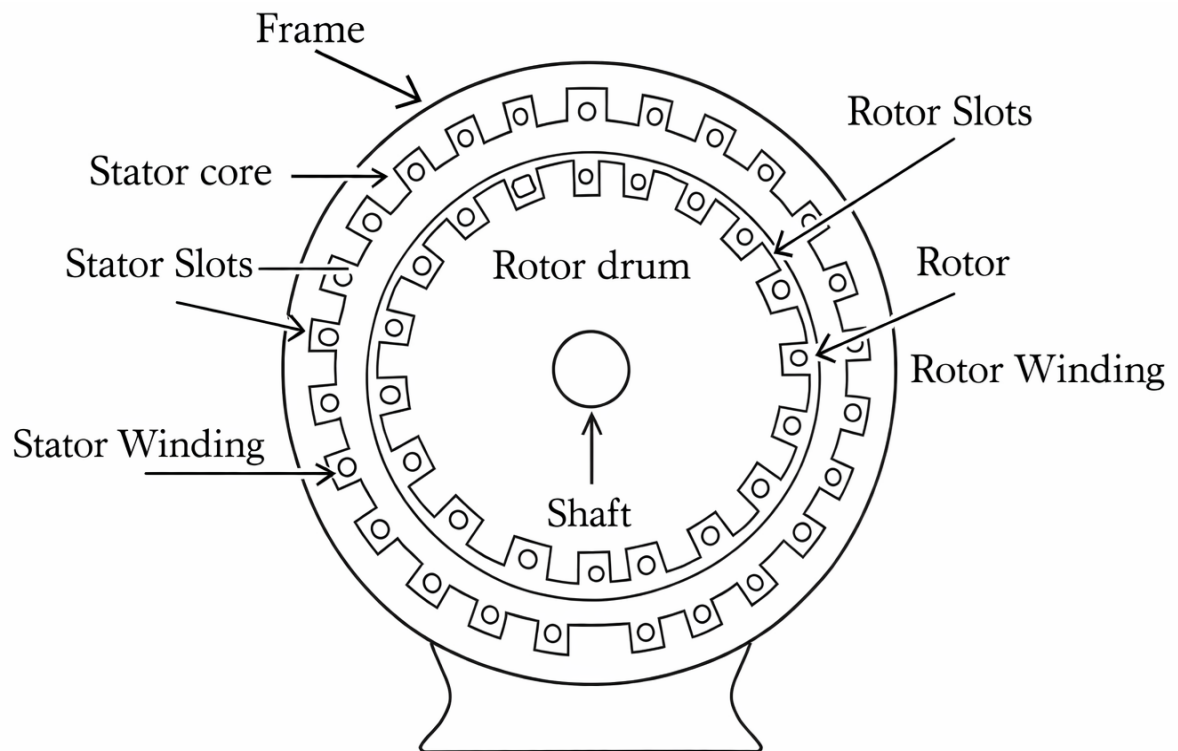
Ans: Construction: A typical motor consists of two parts namely stator and rotor like other type of motors.

Stator construction: The stator of an induction motor is laminated iron core with slots. Coils are placed in the slots to form a three (or) single-phase winding.

1. Stator Frame: It is the outer part of the three-phase induction motor. Its main function is to support the stator core and stator winding. The frame is either made up of die-cast or fabricated steel.

2. Stator Core: The main function of the stator core is to carry alternating flux. In order to reduce the eddy current losses, the stator core is laminated. The lamination sheet are generally made up of silicon steel, which reduces the hysteresis loss.

3. Stator Winding: The slots on the periphery of stator core of the three phase induction motor carries three phase windings. This three phase winding is supplied by three phase ac supply.



Rotor construction: Type of rotors Rotor is of two different types.

1. Squirrel cage rotor
2. Wound rotor (Or) slip ring rotor

Squirrel cage rotor: In the squirrel-cage rotor, the rotor winding consists of single copper or aluminium bars placed in the slots and short-circuited by end-rings on both sides of the rotor. Most of single-phase induction motors have Squirrel-Cage rotor.

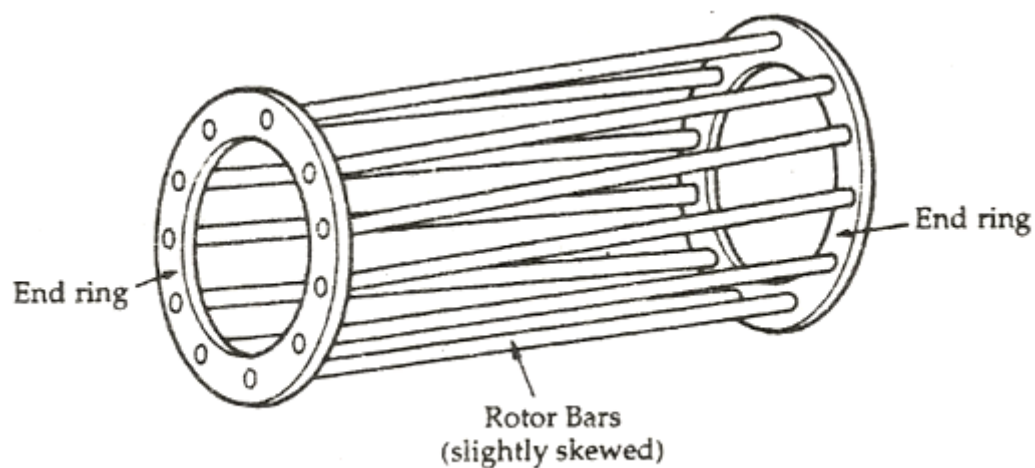


Fig: Squirrel Cage Rotor.

Advantages of Squirrel Cage Induction Motor:

1. Its construction is very simple and rugged.
2. As there are no brushes and slip ring, these motors require less maintenance.

Wound Rotor: In the wound rotor, an insulated 3-phase winding similar to the stator winding wound for the same number of poles as stator, is placed in the rotor slots. The ends of the star-connected rotor winding is connected to a slip ring.

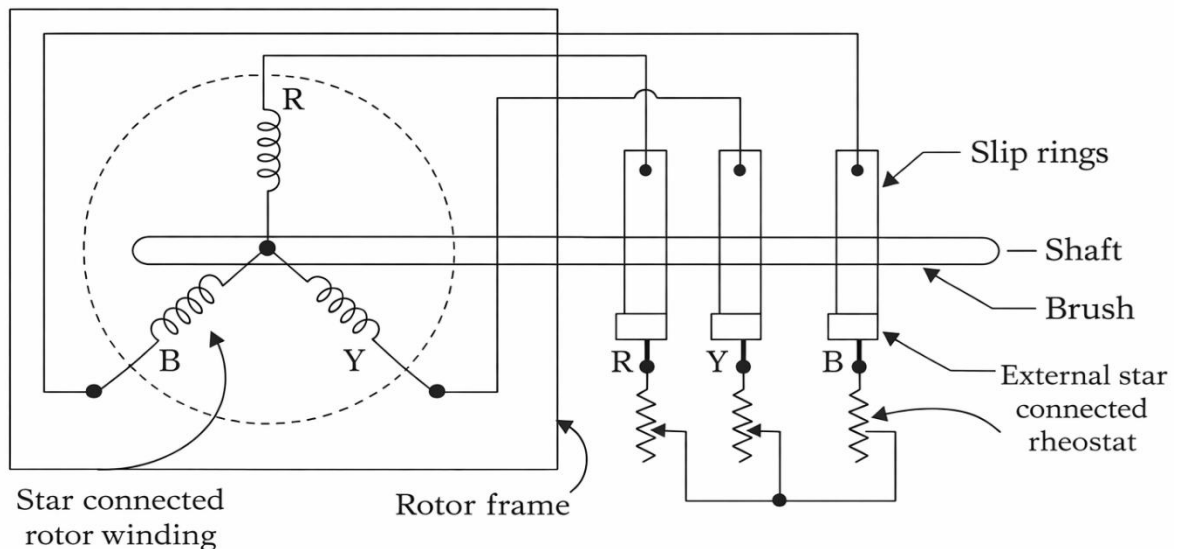


Fig: Slip Ring Rotor.

Advantages of Slip Ring Induction Motor:

1. It has high starting torque and low starting current.
2. Possibility of adding additional resistance to control speed.

7. Explain the principle and operation of three phase Induction Motor.

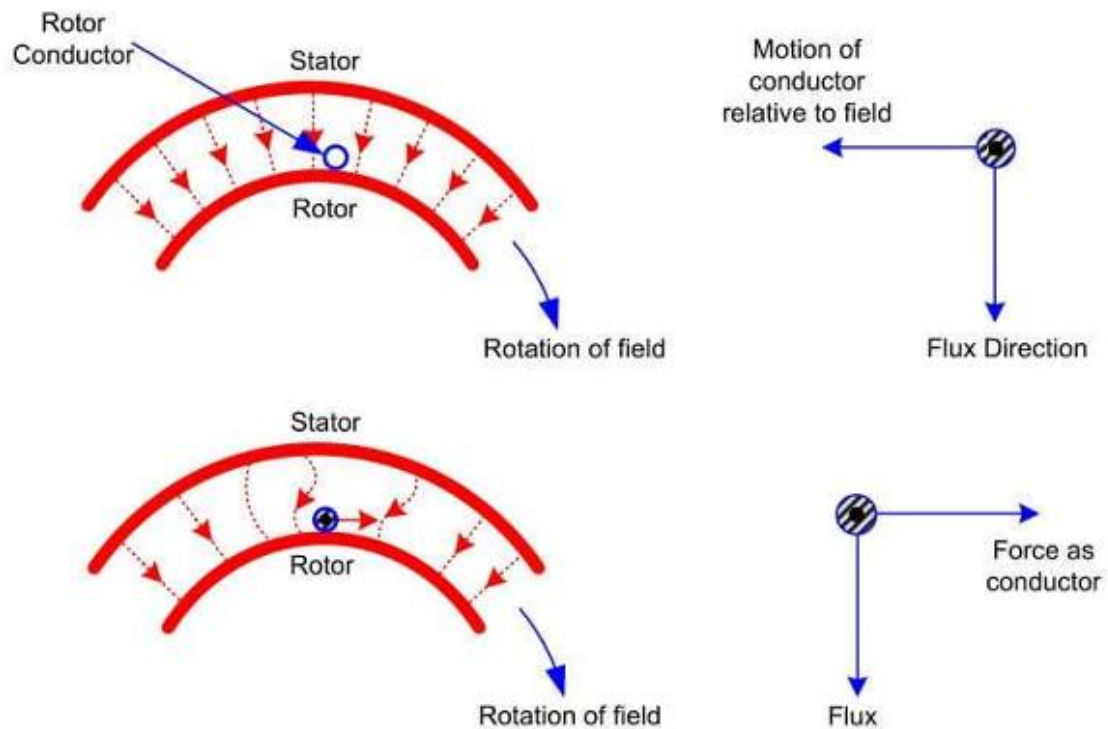
Ans:

Induction Motor Working Principle:

- When a three-phase stator winding of an induction motor is connected to the AC supply, current starts flowing in the stator conductors. Due to this current, a rotating magnetic field (RMF) of constant amplitude and a

constant speed (Synchronous Speed N_s) is set up in the air gap between stator and rotor.

- The rotor winding is still stationary. So rotating magnetic field cuts the stationary rotor conductors and induces an EMF in the rotor winding. Voltage induced in the rotor gives rise to rotor currents.



- The direction of the rotor current is such that it will oppose the cause that produces the current (Lenz's law). And the cause behind producing the rotor current is the 'relative velocity' between the rotor and rotating magnetic field.
- So the rotor current will flow in such a direction that the rotor will experience a force that accelerates it in the same direction as that of the rotating magnetic field. At the no-load rotor rotates at slightly less speed than N_s .

8. Explain the construction of Alternator.

Ans: An alternator has 3-phase winding on the stator and a d.c. field winding on the rotor.

1. **Stator:** It is the stationary part of the machine and is built up of sheet-steel laminations having slots on its inner periphery. A 3-phase winding is placed in these slots and serves as the armature winding of the alternator. The armature winding is always connected in *star and the neutral is connected to ground.
2. **Rotor:** The rotor carries a field winding which is supplied with direct current through two slip rings by a separate d.c. source.
 - (i) Salient (or projecting) pole type
 - (ii) Non-salient (or cylindrical) pole type

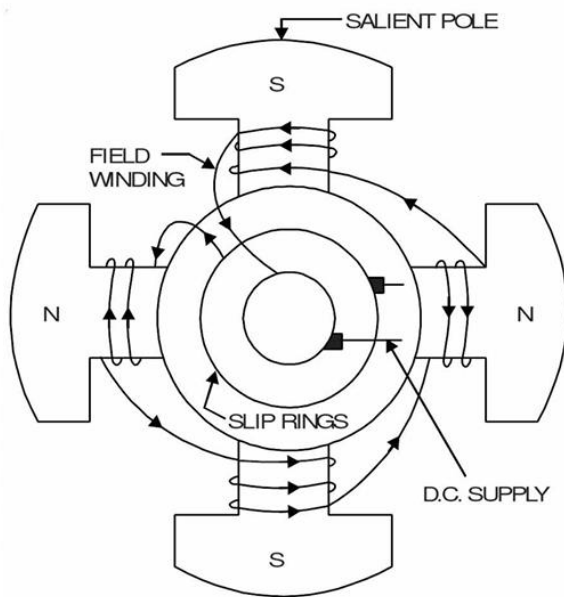


Fig.(a): Salient Pole Rotor

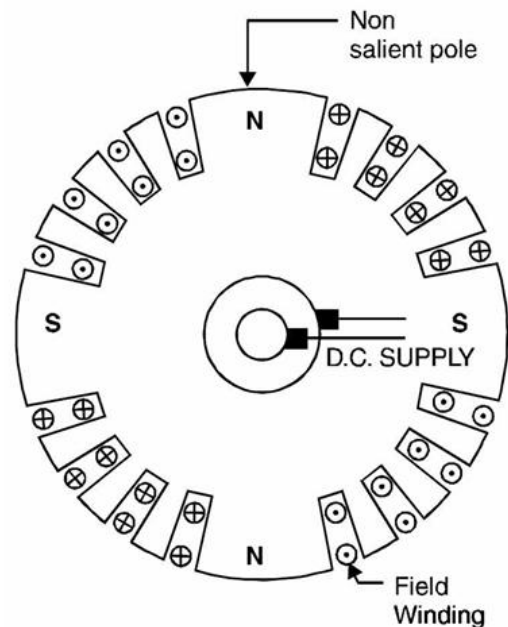


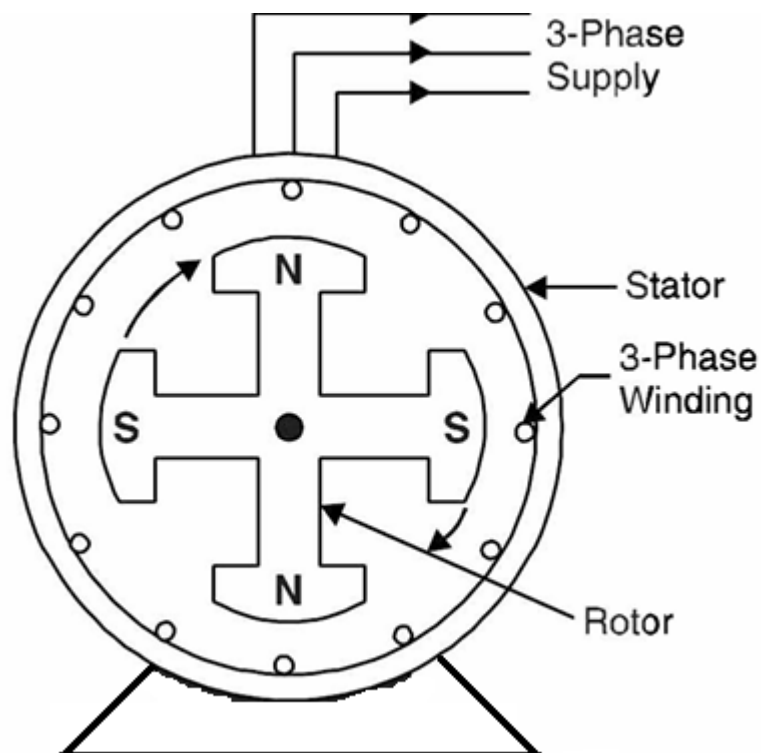
Fig.(b): Non-Salient Pole Rotor

- (i) **Salient pole type.** In this type, salient or projecting poles are mounted on a large circular steel frame which is fixed to the shaft of the alternator as shown in Fig. (a). The individual field pole windings are connected in series in such a way that when the field winding is energised by the d.c. exciter, adjacent poles have opposite polarities.
- (ii) **Non-salient pole type.** In this type, the rotor is made of smooth solid forged-steel radial cylinder having a number of slots along the outer periphery as shown in Fig. (b). The field windings are embedded in

these slots and are connected in series to the slip rings through which they are supplied with direct supply.

9. **Describe the working principle of Alternator.**

Ans: The rotor winding is energised from the d.c. exciter and alternate N and S poles are developed on the rotor. When the rotor is rotated in the clockwise direction by a prime mover, the stator or armature conductors are cut by the magnetic flux of the rotor poles. Consequently, e.m.f. is induced in the armature conductors due to electromagnetic induction.



The induced e.m.f. is alternating since N and S poles of the rotor alternately pass the armature conductors. The direction of induced e.m.f. can be found by Fleming's right-hand rule and the frequency is given by

$$f = \frac{NP}{120}$$

where

N = speed of rotor in r.p.m.

P = number of rotor poles

The magnitude of the voltage induced in each phase depends upon the rotor flux, the number of the conductors in the phase, and the speed of the rotor.

10. Explain the Construction and Operation of PMMC Instrument.

Ans: Construction of PMMC Instrument: The main components include

1. **Permanent Magnet:** A strong U-shaped magnet (Alnico) produces a uniform, stationary magnetic field.
2. **Moving Coil:** A rectangular coil of fine wire wound on a lightweight aluminium frame (former).
3. **Core:** A soft iron cylinder inside the coil reduces reluctance and strengthens the magnetic field.

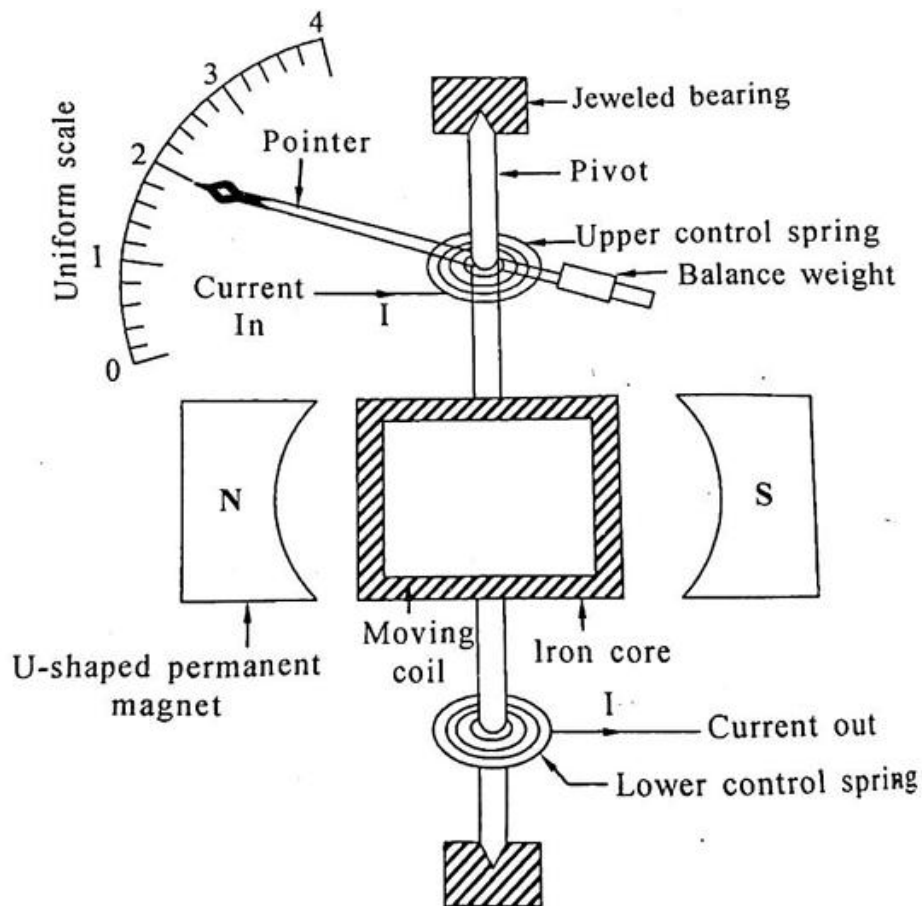


Fig.(a): Permanent magnet moving coil instrument.

4. **Control System:** Two hairsprings (phosphor bronze) provide restoring torque and deliver current to the coil.

5. **Damping System:** Eddy current damping, produced by the movement of the aluminium former in the magnetic field, prevents oscillation.
6. **Pointer and Scale:** A pointer attached to the coil spindle moves over a calibrated linear scale.

Operation of PMMC Instrument:

- When the instrument is connected in the circuit to measure current and voltage, the operating current flows through the coil.
- When the current is passed through the coil, a deflecting torque is produced due to the reaction between permanent magnetic field and the magnetic field of the coil. This deflecting torque is proportional to the value of current in the coil

$$T_d \propto I$$

- Due to this torque, the coil moves and the pointer deflects over the calibrated scale. When the coil moves, the springs are twisted or rewound, thus the spring offers a controlling torque to the moving system.
- The controlling torque due to spring is directly proportional to the deflecting of the moving part $T \propto \theta$. When the deflecting and controlling torque are equal $T_d \propto T_c$ the pointer comes to rest at its final position, hence, $\theta \propto I$.
- Since the deflection θ is directly proportional to the current (I), these instruments have uniform scale.

Deflecting torque: It is directly proportional to the current or the voltage to be measured. So, the instrument can be used to measure direct current and DC voltage.

Control torque: The torque which controls the deflecting torque is called control torque (Spring control). It brings the pointer back to zero position when the deflecting torque is removed.

Damping torque: Eddy current damping. When the moving coil made of aluminium former is moved due to the force exerted on it, it cuts the magnetic

flux lines produced by the permanent magnet. Hence, eddy currents are induced in the former.

11. Discuss the Principle and operation of Moving Iron Instrument with a neat sketch.

Ans: Principle of Operation: The operating principle is based on the force of attraction or repulsion between a stationary coil and a moving iron piece placed in its magnetic field.

- **Attraction Type:** The magnetic field produced by the coil attracts a soft iron piece into the coil.
- **Repulsion Type:** The coil magnetizes two soft iron vanes with the same polarity, causing them to repel each other

Attraction type moving iron instrument: It works on the principle of attraction of a single piece of soft iron towards a magnetic field.

Construction: The Fig (a) shows the construction of an attraction-type moving iron instrument. It has a fixed coil (or) solenoid through which current flows. An oval-shaped soft iron piece is attached to a spindle, allowing it to move in and out of the coil.

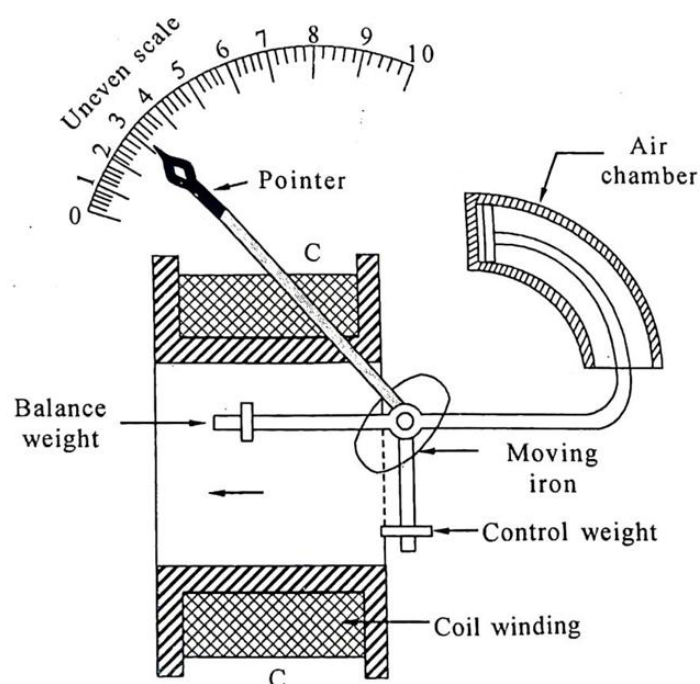


Fig.(a): Attraction type Moving Iron instrument

A pointer is connected to the spindle to indicate readings on a calibrated scale. The spindle is pivoted in jewel bearings at both ends for smooth movement. A spiral spring provides controlling torque, while an aluminium vane, moving in a closed air chamber, provides damping torque.

Working:

- When the instrument is connected in the circuit to measure current or voltage, the operating current flows through the coil.
- Whenever current flows through the coil the electromagnetic field is set up along its axis. In other words, the coil behaves like a magnet and therefore it attracts the soft iron piece towards it, thus providing the deflecting torque T_d
- The deflecting torque causes the pointer attached to the moving system moves from zero position over a calibrated scale. The deflecting torque depends upon the value of magnetic flux (current flowing through the coil).
- The spiral spring offers the controlling torque which depends upon the angle of twist. When the deflecting torque is equal to the controlling torque ($T_d \propto T_c$) the pointer will come to rest at a position.
- The deflecting torque (T_d) is proportional to square of the current I^2 and controlling torque T_c is proportional to angle of twist ' θ '. At a steady position of deflection $T_d \propto T_c$, hence $\theta \propto I^2$. In case of AC measurement, $\theta \propto I_{\text{rms}}^2$.
- Hence, the scale is not uniform, being crowded at the beginning and spread out near the finishing end of the scale.

12.Explain the procedure of finding the value of an unknown resistance using Wheatstone Bridge.

Ans: The Wheatstone Bridge is an instrument designed to measure unknown resistance in electrical circuits. It calculates the unknown resistance by balancing the two legs of the bridge circuit.

Working Principle: Wheatstone Bridge works on the principle of null deflection i.e., there is no current flowing through the galvanometer, and its needle shows no deflection, hence the name null deflection. In the unbalanced

state of the Wheatstone bridge i.e., when the potential across the galvanometer is different, the galvanometer shows the deflection, and as the bridge becomes balanced by changing the variable resistor, the potential difference across the galvanometer becomes zero i.e., the equilibrium state of Wheatstone bridge.

Theory

A Wheatstone bridge consists of four resistances arranged in a bridge form.

Let:

- P and $Q \rightarrow$ Known ratio arms
- $R \rightarrow$ Adjustable known resistance
- $R_x \rightarrow$ Unknown resistance

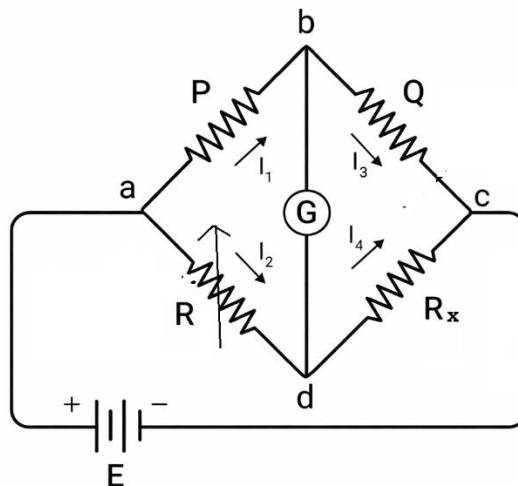


Fig.(a): Wheatstone Bridge

When the bridge is **balanced**, no current flows through the galvanometer (null deflection).

At balance condition:

$$\frac{P}{Q} = \frac{R}{R_x}$$

Therefore,

$$R_x = \frac{Q}{P} \times R$$